SANET: A Shape-Aware Building Footprints Extraction Method in Remote Sensing Images by Integrating Fourier Shape Descriptors

Anna Hu, Liang Wu, Yongyang Xu, and Zhong Xie

***Abstract*— While most state-of-the-art building extraction methods can generate precise binary segmentation masks, geographic and cartographic applications typically require vectorized footprints of the extracted building instead of the rasterized output. Current vectorized footprint extraction methods for multiple buildings have yet to address the confusion brought about by the broken line formed at the slope connection of the building roof. In addition, these methods have some shape errors, such as incomplete and irregular boundaries. Given the above issues, this study proposed a new building outline extraction method, SANET, which uses a transformer block to capture refined boundaries and footprint features. Moreover, a shape-aware loss function was designed to constrain the building shapes and optimize boundary feature generation. Incomplete and irregular boundaries and intersecting outlines aggravate the shape errors for vectorized building objects. Thus, this study computed a Fourier descriptor for a footprint generation model to provide prior shape knowledge for the shape constraint module. Experiments were conducted on the WHU and SpaceNet datasets, and the proposed method could achieve state-of-the-art performance with a higher average precision (AP) and recall compared with other contour-based methods. The proposed shape constraint methods obtained complete and shape-correct building boundaries. Moreover, the incomplete outlines and smooth corners were remarkably improved.**

***Index Terms*— Building footprints extraction, Fourier transform, remote sensing image, shape errors.**

I. Introduction

AS AN essential element in "smart city" construction, the position information of the bottom outline of buildings has become essential data for navigation and positioning. The vectorized position information of building footprints can clearly express the geometric form of a building, which can be crucial to urban geographic information and renewal. Higher requirements have been provided for the precision of building vector coordinates in the current urban development. Building extraction methods are based on dense point cloud data and remote sensing images. Compared with dense point cloud data, remote sensing technology can obtain full-coverage, large-scale, high real-time surface observation data, efficiently collecting large-scale building information. Accurate and automatic extraction of vectorized buildings from high-resolution remote sensing images is crucial to urban planning, map data updating, and emergency response.

Traditional building extraction methods from high-resolution remote sensing images can be divided into morphological index methods [1], [2] and image classification methods [3], [4]. Morphological index methods use building structural elements to measure or extract corresponding shapes or features. However, these methods must consider the diversified building features and are confined to some typical shapes. Image classification includes object- and pixel-oriented methods. Whether at the pixel or object level, the original machine learning methods can only reflect low-level features and cannot manage higher-level semantic information. To consider the high-dimensional semantic information and texture features in remote sensing images, semantic segmentation tasks based on deep learning have been widely used in remote sensing building extraction [5]. With increased geographic information system requirements, a single building shape becomes the unit during urban construction tasks. Therefore, instance segmentation methods have become the primary solution in recent building extraction research [6], [7]. These methods focus on improving extraction accuracy, such as enhancing texture, extracting building location information, and combining contextual features. However, these methods ignore the boundary shape problem and cannot generate footprint information.

The fundamental object of urban spatial vector data research is acquiring vectorized building contour information, which is necessary to study the spatial pattern of buildings and explore urban development. Thus, many researchers have attempted to find a method to obtain vectorized building outlines from optical remote sensing images and proposed the raster building outline conversion method. Initially, the semantic and instance segmentation methods introduced the raster-to-vector conversion methods to generate vector outlines from binary results [8], [9]. The building shape details will be lost in the vector-to-raster conversion process.

Several researchers have tried to use the recurrent neural network (RNN) block to locate the boundary points after obtaining the boundary feature [10], [11]. Considering the large number and various styles of urban buildings, the

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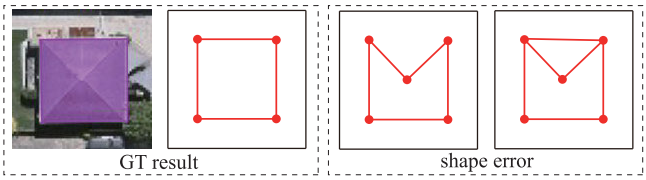


Fig. 1. Shape errors in recent building outline extraction methods.

processing efficiency of single-building extraction methods must be improved. Thus, some research combined instance segmentation methods with the RNN model, which can extract multitarget buildings from remote sensing images [9], [12]. Except for the RNN model, other algorithm designs can obtain vectorized footprints [13], [14], [15]. These methodologies, akin to approaches based on the RNN model, leverage target detection algorithms to extract positional information. Subsequently, contour extraction is performed on the architectural targets localized at the current position. These methods exploited the hierarchical supervision of bottom-level vertices, mid-level line segments, and high-level regional masks to fit vectorized building outlines. Although these methods realize extracted multivector building outlines, shape errors were still found during extraction (see Fig. 1).

Recent building outline extraction methods have used several approaches to obtain vectorized building footprints; however, the shape error in building outlines is ignored. The most critical shape issues are given as follows.

1) Given the increased resolution of the remote sensing dataset, the broken line formed at the slope connection of the building roof is mistakenly detected as the outline. Thus, the inaccurate building boundary generates an incomplete building outline.
2) The shape feature for building shapes must be preserved in these methods, but shape errors (such as incomplete and irregular building boundaries and intersecting outlines) are uncontrollable.

To visually demonstrate the efficacy of the proposed shape-aware building footprint extract methods, this study employs the structurally more straightforward PolyMapper method as a baseline. Given the shape errors in current building outline extraction methods, this study considers the integrity and shape correctness of the building shape. It proposes a vectorized building outline extraction method that optimizes building feature extraction and footprint generation blocks. To address the integrity issue of the building boundary, the transformer block is used to capture refined boundaries and footprint features. Furthermore, a shape-aware loss function is designed to constrain the building shapes to optimize boundary feature generation. For the vectorized footprint generation method, the shape errors considerably affect the validity of the extracted building outline. Thus, this study computes a Fourier descriptor for the footprint generation model, which can obtain shape information.

II. Related Work

Many scholars have used edge extraction, image segmentation, and machine learning to achieve building target detection based on high-resolution remote sensing images. With the increasing demand for building location information applications, the accuracy of building outlines has improved. Moreover, building extraction methods are gradually developing from pixel to vector levels.

Pixel-Level Building Extraction Algorithm: Pixel-level building extraction algorithms are semantic segmentation methods that predict each pixel's category. With the instant development of computer vision methods, semantic segmentation methods have been successfully applied to high-resolution remote sensing image semantic segmentation tasks [16], [17]. CMGFNet [18] extracts multilevel features from RGB and DSM data by using two separate encoders, and this method provides cross-modal and multilevel feature fusion methods to fuse features in two modalities. The method [19] attempts to solve the problem in previously proposed hybrid segmentation methods, optimizing the segmentation and classification stages. This method captures building features from remote sensing images and improves accuracy. Given its outstanding accuracy, the method has been widely used in previous ground object extraction research. However, semantic segmentation methods have more stringent sample labeling requirements, and the number of network layers and parameters does not apply to massive remote sensing images. While some methods try to solve the problem in building outlines, the CBR-Net [20], the methods in [21], and the HRNet [22] enhance the ability of feature extraction that improves the imperfect boundary. Those methods cannot distinguish the closely connected buildings. These methods also influence contextual information, which often appears as a phenomenon where a target is divided into multiple parts or different categories of targets are classified into the same category.

Object-Level Building Extraction Algorithm: To meet the requirements of cartographic and geographic information systems for building data collection and automatic updating, instance segmentation methods have been proposed to distinguish each building outline from multibuilding images. The method in [23] used transfer learning methods on ResNet as the backbone, whereas the feature pyramid network (FPN) architecture was used to extract features. The method in [24] used Mask R-CNN to segment 2-D images and projected the instance masks onto 3-D point clouds to establish 2-D–3-D correspondences. The methods in [25] proposed an instance segmentation model to obtain a typical object class, combining multiple active contour evolution over semantic segmentation maps and fully convolutional networks. These methods combined object detection with semantic segmentation, and the binary object results were final. The object-level building extraction algorithm can better solve the problem of salt-and-pepper noise and maximize the attribute feature information of building objects, thus effectively improving the classification accuracy of high-resolution images. These methods cannot obtain vectorized building outlines from binary results. Thus, the raster-to-vector conversion method was introduced to obtain coordinate information from binary results.

Line-Based Vectorized Building Extraction Methods: These methods are dedicated to extracting the edge lines of buildings using morphological algorithms, field maps, and other

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techniques. Initially, line extraction techniques based on morphological algorithms mainly used the Canny operator [26], Hough transform [27], and LSD method algorithms. However, limited by the effectiveness of morphological algorithms in various complex scenarios, wireframe parsing theories [28], [29], [30] have been designed and widely applied based on neural network algorithms. The algorithm for line extraction based on field maps employs attractive field maps [31], [32] and conditional random fields to represent lines. FrameField [15] adds a frame field output to a deep segmentation model to extract buildings from remote sensing images. The extraction of feature lines has emerged as a prominent challenge in computer vision. Because the required results for building extraction research are complete building polygons, and the line segments obtained from the line extraction algorithm must be connected to form a complete building polygon, the gap between lines and regions must be reduced during the connection process. Solving the topology errors generated while connecting lines to polygons is difficult in building extraction research. Complex optimization techniques for reconstructing plane structures also reduce the efficiency of building extraction.

Contour-Based Vectorized Building Extraction Methods: Many researchers use deep learning methods to generate vectorized building outlines directly, thus avoiding the error of the vector–raster conversion results. The semiautomatic data annotation method, PolyRNN [10], was initially proposed to extract object outlines from manual annotation. PolyRNN introduces an RNN block to detect footprints coordinated by one-hot encoding. Relying on the RNN block, these methods generate more compact and regularized representations that solve the problem in rasterized results. The RNN block uses the one-hot encoder method to locate the building footprints, and it proposes a novel sequence-based algorithm that defines the building outline as a closed polygon. Substantial research used PolyRNN as the baseline network [33], [34] and improved the structure to obtain more precise object results. The Polygon-RNN-PP [11] model is the optimization algorithm of Polygon-RNN, which introduces the attention network and evaluator network, and a gated graph neural network is used to increase the resolution. However, these single-object extraction methods cannot be applied to remote sensing images containing multiple buildings in a single image. Thus, researchers combined object detection methods with RNN blocks to design a multivector building extraction structure [9]. These methods focus on multiobject extraction and obtain regular building boundaries. Based on the PolyMapper method, several methods have been used to extract vectorized building contours from rasterized remote sensing images [13], [35]. HiSup [14] exploited the hierarchical supervision of bottom-level vertices, mid-level line segments, and high-level regional masks to fit vectorized building outlines. PolyWord [36] uses a graph neural network to predict the connection strength between each pair of vertices and estimates the assignments by a differentiable optimal transport. These studies proposed several methods to generate vectorized footprints and obtain building outlines by connecting footprints in sequence. However, these methods do not focus on shape accuracy in the global algorithm; shape errors are shared in those results.

III. Methodology

Based on the original vectorized building outline extraction model, this study proposed a shape-aware building outline extraction method, SANET. This method focuses on the problem of inaccurate boundaries and shape errors and considers the requirements to process multibuilding outlines in single remote sensing images in the structure. Thus, SANET introduces FPN and region proposal network (RPN) blocks. The FPN block generates the multidimensional feature to extract boundary and vertex features, whereas the RPN block extracts the possible anchor frame from the multidimensional feature. In addition to these baseline networks, the shape integrity retention and shape constraint modules were designed. The shape integrity retention module used a transformer block to capture refined boundaries and footprint features. Furthermore, a shape-aware loss function was designed to constrain the building shapes and optimize boundary feature generation. The shape constraint module computed a Fourier descriptor for the footprint generation model, which can provide prior shape knowledge. The structure of SANET is shown in Fig. 2.

A. Shape Integrity Retention Module

Given the increased resolution of the remote sensing dataset, the broken line formed at the slope connection of the building roof is mistakenly detected as the outline. Thus, the inaccurate building boundary generates an incomplete building outline. As the most critical shape constraint for generating the position of footprints, the building boundary and vertex information should be emphasized. To address the issue of the integrity of the building shape, the transformer block is designed to capture refined boundaries and footprint features. The robust feature extraction ability can considerably avoid the misjudgment of the ridgeline. The transformer block comprises an encoder and a decoder block. The encoder models the context of semantic concepts in multiple features of FPN, whereas the decoder projects the corresponding semantic information (boundary and vertex).

In the building outline, footprints with the same semantic characteristics may show the same spectral characteristics. The convolution kernel is translation invariance and local sensitivity, which can capture local spatial information. However, the convolution needs a global understanding of the image and cannot obtain the context information in high-dimensional features. The nonlocal self-attention method performs well by modeling the dense relationship between pixels. Therefore, this study attempts to replace the original convolutional operation and migrate the transformer model to capture global shape information in buildings. The transformer model can excavate long-distance dependencies and learn the most appropriate inductive bias according to different task objectives. The transform block divides the image into square grids, flattens each square into a single vector by connecting all pixel channels in a square, and then linearly projects them to the required input dimensions.

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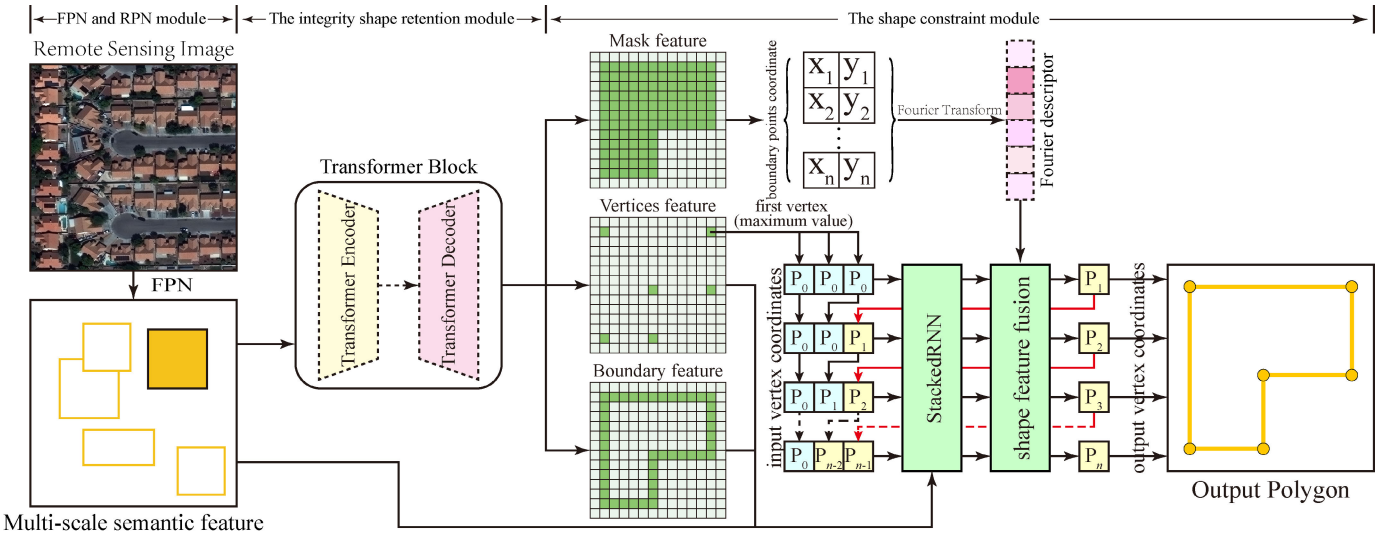


Fig. 2. Structure of SANET.

To retain the integrity of the building shape, this study designed the feature integration transformer block to capture the global spatial building footprint information. The structure of the feature integration transformer block is shown in Fig. 3.

To preserve the boundary and footprint features, the shape integrity retention module used multisemantic features as the initial semantic information generated by the FPN block. The transformer block consists of the semantic token-generated block, the transformer encoding block, and the transformer decoding block. The semantic token-generated block is used to convert the original feature information to a group-compact semantic token, which can satisfy the input of transformer encoding. The boundary and vertex information can be integrated into a semantic token. The transformer encoding block extracts spatial features and models global semantic information. In this study, the transformer encoding block used four transformer blocks to extract spatial features. The transformer block can fully utilize token-based global semantic relationships comprising layer normalization, multihead attention, residual structure, and MLP blocks. The multihead attention block can jointly receive information from different presentation subspaces processed at different locations. The generated encoding feature with rich global information is input into the continuous vectorized footprint-generated block as global spatial semantic information. The context-decadent tokens generated from the transformer encoder contain compact high-level semantic information that can reveal interest change. The transformer decoder must project the representation of concepts back to pixel space to obtain pixel-level features. To achieve this, the transformer decoder refines the image features of each temporal.

B. Shape Constraint Module

The shape errors considerably affect the validity of the extract building outline for the vectorized footprint coordinate generation method. The shape feature of building shapes must be preserved in current building outline extraction methods, and shape errors, such as incomplete and irregular building boundaries and intersecting outlines, are uncontrollable. Combining the efficient shape-aware feature with shape space knowledge can effectively enhance the accuracy of building outline extraction.

The building shape feature vectors should be simple but highly recognized in the building vector footprint extraction model. The Fourier shape descriptor satisfies the above two conditions, and the vector length is independent of the number of footprints of the building polygon. Accuracy needs can also be adjusted. Thus, this study uses a Fourier descriptor for the footprint generation model, which focuses on the shape of the building outline and provides building shape knowledge that conforms to the geometric shape relationships. The Fourier shape descriptor with shape feature parameters is introduced into the LSTM block as a shape feature vector, which can effectively constrain the generation of footprint position information. This module can translate location information for the Fourier descriptor so that the shape error can be restricted. The sketch map of the Fourier descriptor vector is shown in Fig. 4.

Because the calculation object of Fourier shape descriptors is the grid contour of buildings, but the grid contour of the building is not accurate during the training process, how to combine the calculation process of Fourier shape descriptors with the training has become the primary issue in integrating morphological algorithms and deep learning models. Therefore, this study adopts the strategy of inconsistent training and testing processes, which is commonly used in current deep learning models. During the training process, the Fourier shape descriptor algorithm takes real building entity contours as the object of computation. The geometric shape knowledge of real building entity contours will be integrated into the building extraction model during the training process. During the model testing process, the contour features in this method are used to obtain the Fourier shape descriptor. At this point, the trained building extraction model can capture prior knowledge of building shape and constrain the generation process of building footprints.

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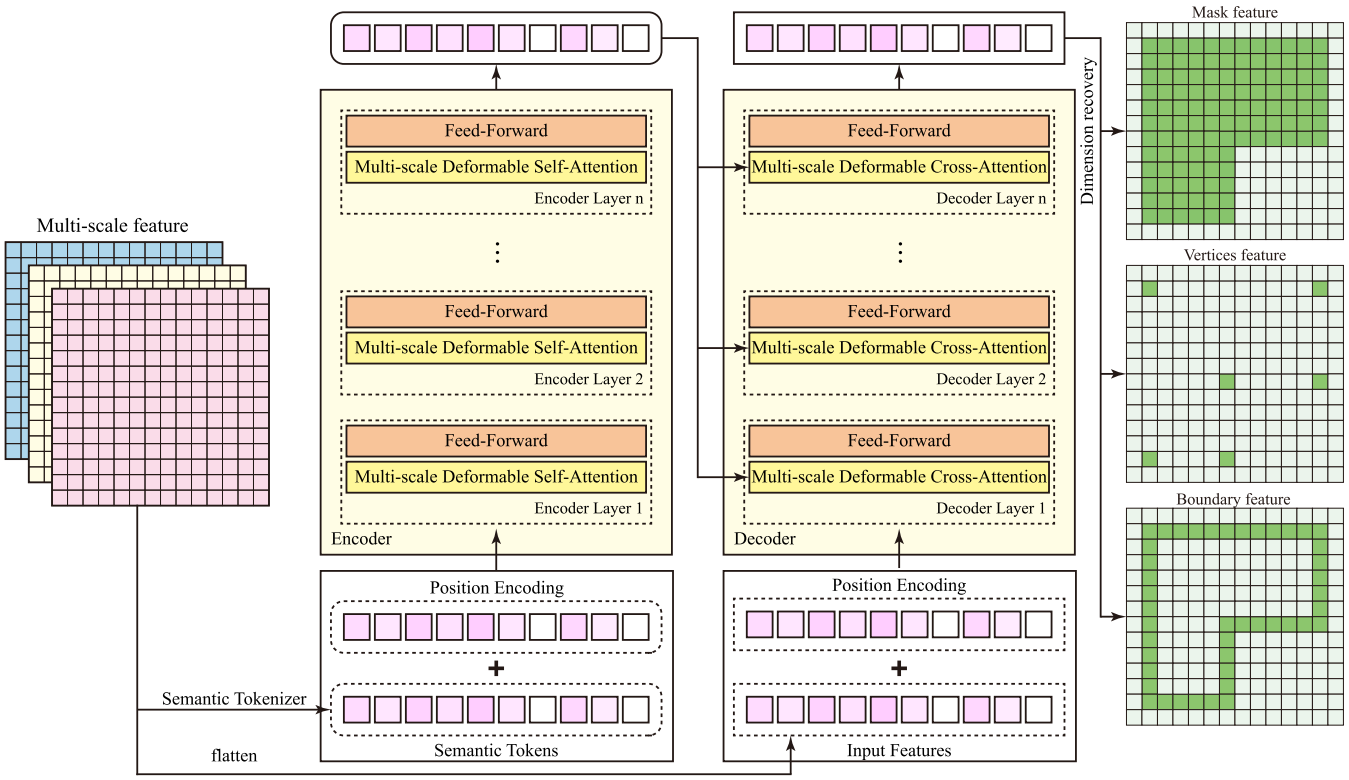


Fig. 3. Structure of feature integration transformer block.

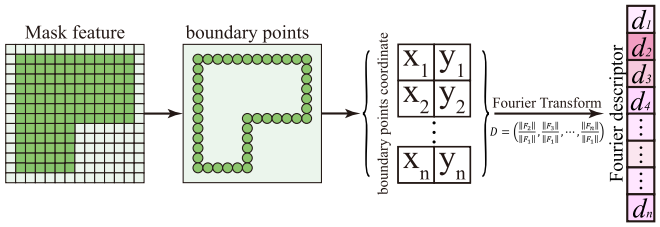


Fig. 4. Fourier descriptor generated methods.

The building polygon comprises a set of coordinates with overlapping start and endpoints. In the plane right-angle coordinates, arbitrary points are taken as the initial points $(P_0(x_0, y_0))$, and points are defined clockwise $(P_1, P_2, \ldots, P_n)$ to constitute a closed building boundary (P_0 and P_n are the same points).

Introducing the elements in the plane right angle to the complex plane can obtain the coordinate expression of the points, and $P(l)$ is the periodic function with building perimeter L as the period. Thus, the Fourier series can describe the building footprint information, and all coefficients F_n of the Fourier series are complex numbers. The modulus vector of F_n forms the modulus vector F of the Fourier series as the following equation:

$$F = (\|F_1\|, \|F_2\|, \ldots, \|F_n\|). \tag{1}$$

The modulus vector F of the Fourier series was normalized to make it invariant to rotation, scale, and shift. Each term in the modulus vector F was divided by the first term (modulus of coefficient F_1). Given that the first term after processing is 1, the Fourier shape descriptor D is obtained by dividing the first term as the following equation:

$$D = \left(\frac{\|F_2\|}{\|F_1\|}, \frac{\|F_3\|}{\|F_1\|}, \ldots, \frac{\|F_n\|}{\|F_1\|}\right) = (d_1, d_2, \ldots, d_{n-1}). \tag{2}$$

The length K of the Fourier descriptor D tends toward infinity, and judgments should be made based on accuracy requirements in generating descriptors. In this study, the length of the Fourier descriptor is set up by the overlap between the polygon restored by the Fourier shape descriptor and the original building polygon. The specific experimental process is explained in the discussion.

The footprint generation methods consider the computational complexity and the regularized building boundary feature and use an LSTM block to enhance the selected point strategy. This algorithm comprises some LSTM time steps and generates footprint position information. Each LSTM cell in the time step predicts the subsequent footprint's position information using the first two and the head footprints. The head footprints are used to discriminate whether the building outline is closed. The input of each time step is concatenated by the boundary feature, vertex feature, and three footprints. On the one hand, the boundary and footprint features are calculated by the transformer block and preserve the integral building shape information, which is constrained by shape-aware loss. On the other hand, the three footprints represent the head footprints and the first two of the current footprints. The head footprints determine the beginning and end coordinates of the building, and the two footprints can uniquely determine the direction for the next coordinate. The structure of the LSTM block is shown in Fig. 5.

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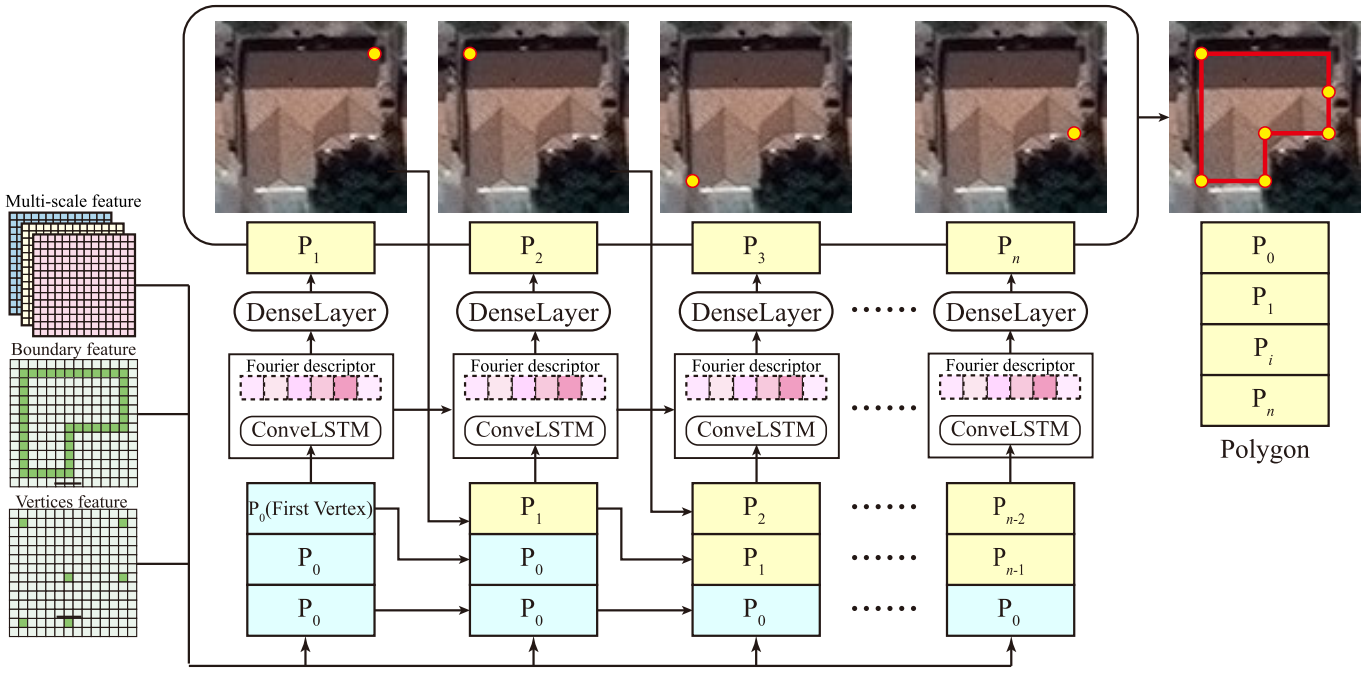


Fig. 5. LSTM structure of the footprints predicting the method.

C. Loss Function

In this study, the loss function comprises shape-aware loss and footprint-generated loss. The shape-aware loss constraints single building shape by optimizing footprint ($L_{\text{footprint}}$), boundary (L_{boundary}), and mask (L_{mask}) features. $L_{\text{footprint}}$ and L_{boundary} are the cross-entropy losses.

The footprint loss function is the error between the vertices feature obtained through the shape integrity retention module and the label of the building. The footprint loss function can be expressed as follows:

$$L_{\text{footprint}} = -\frac{1}{N}\sum_{i,j}\left(F_{i,j}\log\left(\hat{F}_{i,j}\right) + \left(1-F_{i,j}\right)\log\left(1-\hat{F}_{i,j}\right)\right) \quad (3)$$

where F is the predicted vertical features of the building location and $\hat{F}$ is the label of the building footprint position. N represents the total number of pixels in the feature map.

Similar to the corner loss function, the boundary loss function can be expressed as follows:

$$L_{\text{boundary}} = -\frac{1}{N}\sum_{i,j}\left(B_{i,j}\log\left(\hat{B}_{i,j}\right) + \left(1-B_{i,j}\right)\log\left(1-\hat{B}_{i,j}\right)\right) \quad (4)$$

where B is the predicted boundary features of the building and $\hat{B}$ is the label of the building boundary position. N represents the total number of pixels in the feature map.

To focus on building boundary details, this method treats the pixels around the building boundary as hard-to-classify samples. The illustration of the hard-to-distinguish points is shown in Fig. 6. L_{mask} draws inspiration from using weights in the focal loss to address the issue of sample imbalance and applies spatial weighting in the cross entropy to assign a higher weight to the pixels around the boundary. This study

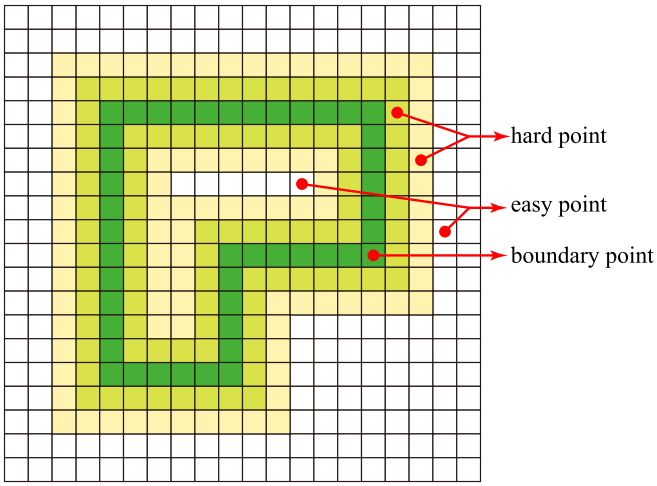


Fig. 6. Illustration of hard-to-classify samples in the building outline.

uses the dilation building boundary minus the erosion building boundary to locate hard-to-classify samples.

Most pixels are located inside the building or background and far from the building boundary; those pixels are the easy points. The points closer to the boundary are more challenging to distinguish; those pixels are the hard points. In contrast to certain boundary operators, this method can produce more substantial object contours and obtain more considerable error rates. The spatial weight maps have been used in cross-entropy loss, which can give high weight to emphasize pixels near the salient boundary. The weight of L_{mask} is shown as follows:

$$M^c = \text{Guass}\left(k * \left((Y,S)^+ - (Y,S)^-\right)\right) + 1 \quad (5)$$

where the Guass(x) is the Gaussian function with the size of 5 × 5, which is used to give a specific weight value

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Fig. 7. Examples of building contours generated in the WHU dataset.

to the area inside the building that does not belong to the boundary, S is the pixel number of dilate and erode process, and K is the hyperparameter. "1" is a matrix with a value of 1 corresponding to the feature size, which sets the pixels of the nonboundary to 1.

The complete L_{mask} is shown as follows:

$$L_{\text{mask}} = -\sum_{x,y} M^{c}_{x,y} * \left(Y_{x,y} * \log Y^{\sim}_{x,y} + \left(1 - Y_{x,y} * \log\left(1 - Y^{\sim}_{x,y}\right)\right)\right) \quad (6)$$

where $Y_{X,Y}$ is the label of the building, $Y^{\sim}_{x,y}$ is the prediction results of the building, and (x, y) is the pixel in the building.

IV. Experiments

A. Datasets

To demonstrate the effectiveness of this article, we have consulted many papers on extracting buildings based on remote sensing images [37], [38], [39], and we have decided to use the WHU dataset [40] with high utilization as the validation dataset. Second, due to the lack of addressing issues such as building shadows, a publicly available dataset with precise building contours from remote sensing images was used. To further validate the effectiveness of this study in complex remote sensing scenes, this study used the SpaceNet dataset [41] with lower clarity and higher vegetation coverage. These datasets consist of high-resolution orthorectified remote sensing images, and buildings are low and densely distributed targets with little height difference.

The WHU dataset comprises aerial and satellite imagery, spanning an area of 1000 km^2, complete with rasterized labels and vectorized maps. The images mainly encompass the building region at the heart of the aerial dataset. To conform to the network input size and alleviate computational pressure on hardware, a remote sensing image with a ground resolution of 0.075 m was subsampled at a 0.3-m ground resolution. The original comprehensive image was then cropped to 256 × 256 pixels. The training and testing datasets encompass 19 221 and 9794 remote sensing images, respectively.

The SpaceNet dataset comprises visible light remote sensing imagery obtained from the WorldView-3 satellite, offering a spatial resolution of 0.3 m. This study used the dataset in Las Vegas, keeping experimental efficiency in mind. Each annotation of building boundaries is a polygon based on a sequence of vertices, i.e., the most classical vector data format. In the SpaceNet dataset, the training set includes 3390 images and 104 128 annotated building footprints, whereas the test set comprises 226 images. All remote sensing images in this dataset have a dimension of 256 × 256. This size ensures the inclusion of sufficient building objects within a single image. The training and testing datasets hold 3390 and 226 remote sensing images, respectively.

B. Evaluation Metrics

This study assessed and contrasted the findings using detection metrics, such as intersection over union (IoU), MS COCO [42], average precision (AP), and average recall

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(AR). Precision and recall scores, given IoU thresholds of 75, were also incorporated in the quantitative evaluation results, denoted as AP75 and AR75. Moreover, a footprint distance metric (FDM) was introduced to highlight the importance of precise building contours [43]. The FDM is the improvement of the PoLiS, which introduces an evaluation score during the match of the original object with the prediction object, and this metric can statistic the unmatched number of target objects. This metric considers locational and morphological variances by interpreting the building boundary as a chain of interlinked perimeters instead of a mere collection of points. In addition to demonstrating superior performance in assessing the congruence of footprints, the FDM was equipped to treat situations where the footprint quantities vary across different buildings, and it can be expressed as

$$p(A, B) = \frac{1}{2m} \sum_{a_i \in A}^{\min} \|a_i - b\| \quad (7)$$

where m is building A's footprint number, $a_(i)$ is building A's footprint, b is the closest point, and ∂B is the point set of footprints and vertical points.

Similarly, $p(B, A)$ is the FDM of building B to building A, and the algorithm is described as follows:

$$p(B, A) = \frac{1}{2n} \sum_{b_j \in B}^{\min} \|b_j - a\| \quad (8)$$

where n is building B's footprint number, b_j is building B's footprint, a is the closest point, and ∂A is the point set of footprints and vertical points.

C. Experimental Setup

This study focused on shape errors in building shapes and proposed a new building outline extraction method. This section shows the building outline results in a large area. Moreover, this experiment compared our methods with the advanced mask-based and the current vectorized building footprint extraction methods. The quantitative and qualitative evaluation results show the results in the WHU and SpaceNet datasets. The results in the tree blocking area and shape error area are all shown in Figs. 8–12.

To demonstrate this study's effectiveness in complex remote sensing scenes, the overall effect in the WHU dataset is shown. Fig. 7 shows the building extraction results of our method in the WHU dataset. From the overall rendering, this study can extract the building outline with the region more comprehensively, with fewer issues of missed and false detections. Moreover, it has excellent generalization for building outline extraction in large-scale scenes.

To present the extraction process of this study holistically across diverse building types in intricate scenes, Fig. 7 further elucidates the extraction outcomes for various architectural styles across a large area. The proposed methodology consistently and accurately extracts their contours.

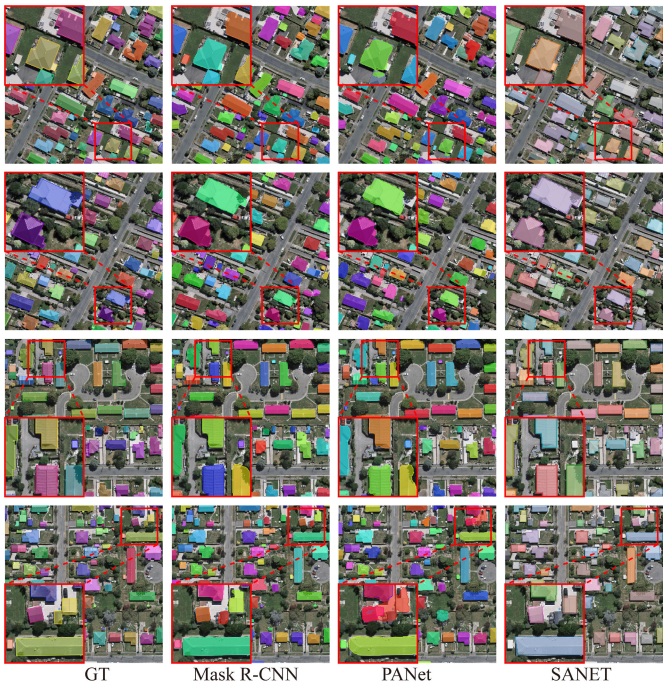


Fig. 8. Example of qualitative building mapping results of mask-based methods (WHU dataset).

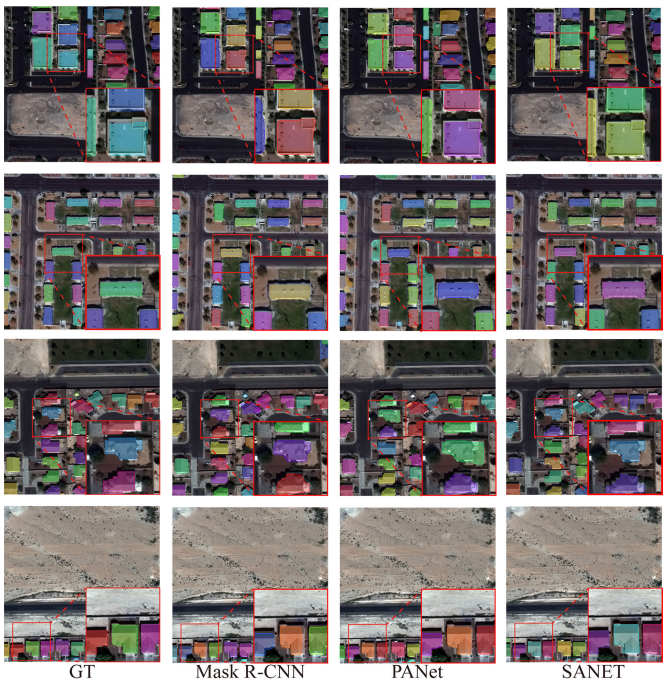


Fig. 9. Example of qualitative building mapping results of mask-based methods (SpaceNet dataset).

The WHU and SpaceNet datasets were used to evaluate the state-of-the-art building boundary extraction methods. The results of vector-based methods (PolyMapper [12], HiSup [14], and Frame-Field [15]) and mask-based methods (PANet [44] and Mask R-CNN) are displayed separately. These methods can all obtain vectorized building footprint results by using different algorithms. To demonstrate the advantages of this method in terms of extracting the shape feature of buildings, this study shows the details of building footprints in the boundary. Tables I and II list the quantitative comparison of

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TABLE I
COMPARISON OF EVALUATION RESULTS ON THE WHU DATASET. THE HIGHEST SCORES ARE IN BOLD

Methods	AP	AP50	AP75	AR	IoU	FDM
Mask-based:						
Mask R-CNN	0.400	**0.695**	0.433	**0.569**	0.821	1.978
PANet	0.368	0.631	0.403	0.519	0.604	2.288
Vector-based:						
PolyMapper	0.225	0.433	0.217	0.324	0.787	2.173
Frame-Field	0.336	0.553	0.364	0.507	0.799	2.009
HiSup	0.297	0.527	0.296	0.374	0.793	2.045
SANET	**0.440**	0.659	**0.512**	0.503	**0.842**	**1.609**

TABLE II
COMPARISON OF EVALUATION RESULTS ON THE SPACENET DATASET. THE HIGHEST SCORES ARE IN BOLD

Methods	AP	AP50	AP75	AR	IoU	FDM
Mask-based:						
Mask R-CNN	0.271	0.527	0.263	0.196	0.781	3.630
PANet	0.252	0.488	0.236	0.434	0.776	3.687
Vector-based:						
PolyMapper	0.341	0.590	0.365	0.419	0.802	3.618
Frame-Field	0.160	0.326	0.142	0.278	0.740	3.664
HiSup	0.230	0.484	0.195	0.314	0.752	3.705
SANET	**0.391**	**0.660**	**0.434**	**0.458**	**0.808**	**3.454**



Fig. 10. Example of qualitative polygonal mapping results of vector-based methods (WHU dataset).

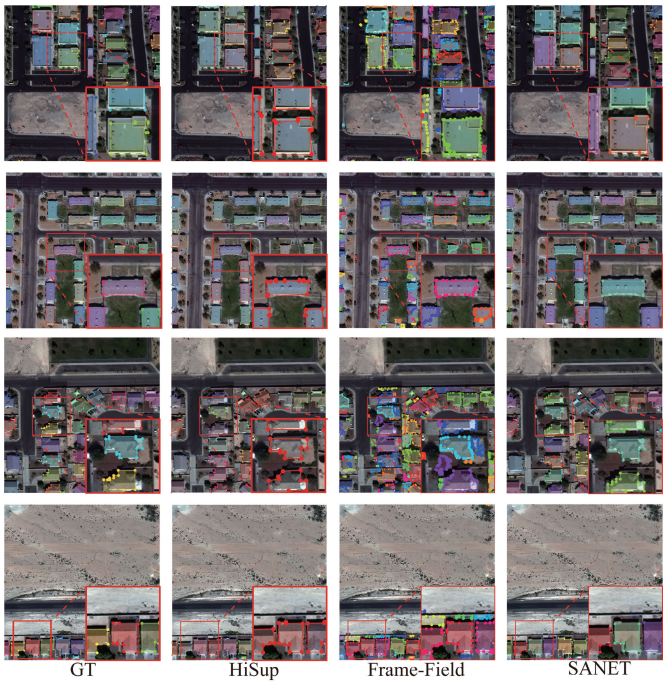


Fig. 11. Example of qualitative polygonal mapping results of vector-based methods (SpaceNet dataset).

mask- and vector-based methods; the AP, AR, FDM, and IoU scores are also displayed.

The quantitative evaluation results show that this study obtained optimal results in almost all evaluated indicators. In the WHU dataset, the scores of AP50 and AR in this study are lower than Mask R-CNN. To further demonstrate the superiority of this study, the details are shown in Fig. 8.

In Figs. 8 and 9, the building outline is displayed in different colors. This study's proposed shape constraint method has a good constraint effect on generating building outlines. The corners and boundaries that conform to the cognitive rules of building shape are presented in the results of this study. Compared with Mask R-CNN and PANet, this study's proposed method focuses on extracting more detailed building boundaries, and the problem of uneven boundary lines and protruding areas that do not conform to the perception of building shape is well resolved. The outstanding boundary information further validates the ability to obtain building shape features. The SpaceNet dataset has a more complex and densely distributed shape of buildings than the WHU dataset,

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Fig. 12. Building outline results in the vegetation occlusion area.

TABLE III
QUANTITATIVE EVALUATION RESULTS OF THE ABLATION EXPERIMENT ON THE WHU DATASET

Methods	AP	AP50	AP75	AR	IoU	FDM
Baseline	0.225	0.433	0.217	0.324	0.787	2.173
Baseline+T	0.413	0.621	0.473	0.485	**0.844**	1.877
Baseline+T+F	0.418	0.630	0.480	0.491	0.841	1.626
Baseline+T+F+S(SANET)	**0.440**	**0.659**	**0.512**	**0.503**	0.842	**1.609**

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF THE ABLATION EXPERIMENT ON THE SPACENET DATASET

Methods	AP	AP50	AP75	AR	IoU	FDM
Baseline	0.341	0.590	0.365	0.419	0.802	3.618
Baseline+T	0.373	0.655	0.398	0.443	0.799	3.624
Baseline+T+F	0.388	0.660	0.432	0.454	0.806	3.495
Baseline+T+F+S (SANET)	**0.391**	**0.660**	**0.434**	**0.458**	**0.808**	**3.454**

and most buildings are covered by vegetation. Therefore, the shape errors in other methods in the SpaceNet dataset are more evident; nevertheless, the method in this study effectively solved these shape problems to a certain extent.

The results of vector-based methods are shown in Figs. 10 and 11. The buildings are composed of footprints and boundaries in different colors, which can clearly show the number of each building.

These results reveal several shape errors in other vector-based methods, which have complex boundary shapes and some indistinguishable broken lines in the roof. The redundant building footprints are standard in the results of Frame-Field and HiSup, but our method can only extract key footprints and maintain precise buildings. This study effectively solved these shape errors. The extracted building outline resulted in relatively complete shapes and considerably reduced the

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Fig. 13. Example of the ablation experiment of building outlines on the WHU dataset.

boundary line crossing phenomenon in the results of other methods. Therefore, this study's proposed shape constraint method has a good constraint effect on generating building outlines.

The distribution of buildings in the SpaceNet and WHU datasets is relatively dense due to trees around the buildings. The vegetation caused the obscure building outlines. In building outline annotation, people learn the building contour in the current remote sensing scene and then predict the building outline under vegetation occlusion. Reasonable prediction of building outlines under occlusion is key to footprint extraction methods. To further demonstrate our method's predictive ability for building outlines in current remote sensing scenarios, the specific details at the occlusion are displayed in Fig. 12.

As shown in Fig. 12, this study considers prior knowledge of building boundary shapes that can better predict the building boundary under occlusion. A raster-to-vector algorithm generates the footprint information of Mask R-CNN and PANet; the dense footprints form a thicker boundary. Although the building extraction results of this study in an occlusion area are incomplete with the ground-truth label, the overall shape of the predicted building edges is better than other methods. The results of this study conform to the primary judgment of building shape in the current scene and are more in line with the annotation personnel's cognition of building shape in ground truth.

V. Discussion

The ablation experiment verifies the functions of the different stages on the WHU and SpaceNet datasets. The baseline methods (PolyMapper), the transformer block, the Fourier descriptor, and the shape-aware loss function are introduced to the baseline model to evaluate the efficiency of this method. "T" represents the transformer block, "F" represents the Fourier descriptor, "S" represents the shape-aware loss function, and "baseline + T + F + S" denotes this study. The extracted results of the ablation experiment are shown in Figs. 13 and 14 and Tables III and IV.

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Fig. 14. Example of the ablation experiment of building outlines on the SpaceNet dataset.

The findings of the quantitative evaluations are delineated in Tables III and IV. The considerable enhancement in the metric of footprint distance confirms that the results acquired in the current research exhibit a higher degree of shape similarity.

In contrast to the baseline model, this study considers the shape features and focuses on optimizing shape errors. The transformer block essentially enhances the efficiency of PolyMapper methods, significantly improving the building extract's accuracy. As illustrated in Figs. 11 and 12, the issue about shapes was substantially eased, addressing the issue of incomplete building shapes resulting from the intersection of boundary lines.

The results of each method reveal that the transformer block, the Fourier descriptor, and the shape-aware loss function have corresponding effects. The transformer block provided more details in overall shape, which fixed smooth building boundaries in the baseline method. The "baseline + T + F" results showed outstanding shape recovery ability, and the difficulty of determining building corners was also addressed. Meanwhile, the shape-aware loss function performed well in restoring the details of the building outlines.

To solve the problem of shape errors, this study designed a Fourier descriptor to calculate the global shape information. Leveraging the inherent attributes of the Fourier transformation, the Fourier descriptor approximates the building outline with varying degrees of precision. Furthermore, these coefficients encapsulate shape-related information, demonstrating invariance not only to translational modifications but also to rotational adjustments and scalar transformations. This encapsulation affords a robust and reliable methodology for processing and interpreting shape-related data. The number of the Fourier descriptor vector is an uncertain value. The preceding vector corresponds to low-frequency information, describing the contour information of the shape, whereas the following vector corresponds to high-frequency information, describing the details of the shape. In shape measurement, the vector of a certain length in front of the Fourier shape descriptor can be retained by actual accuracy needs, and the subsequent details can be discarded. Thus, to determine the

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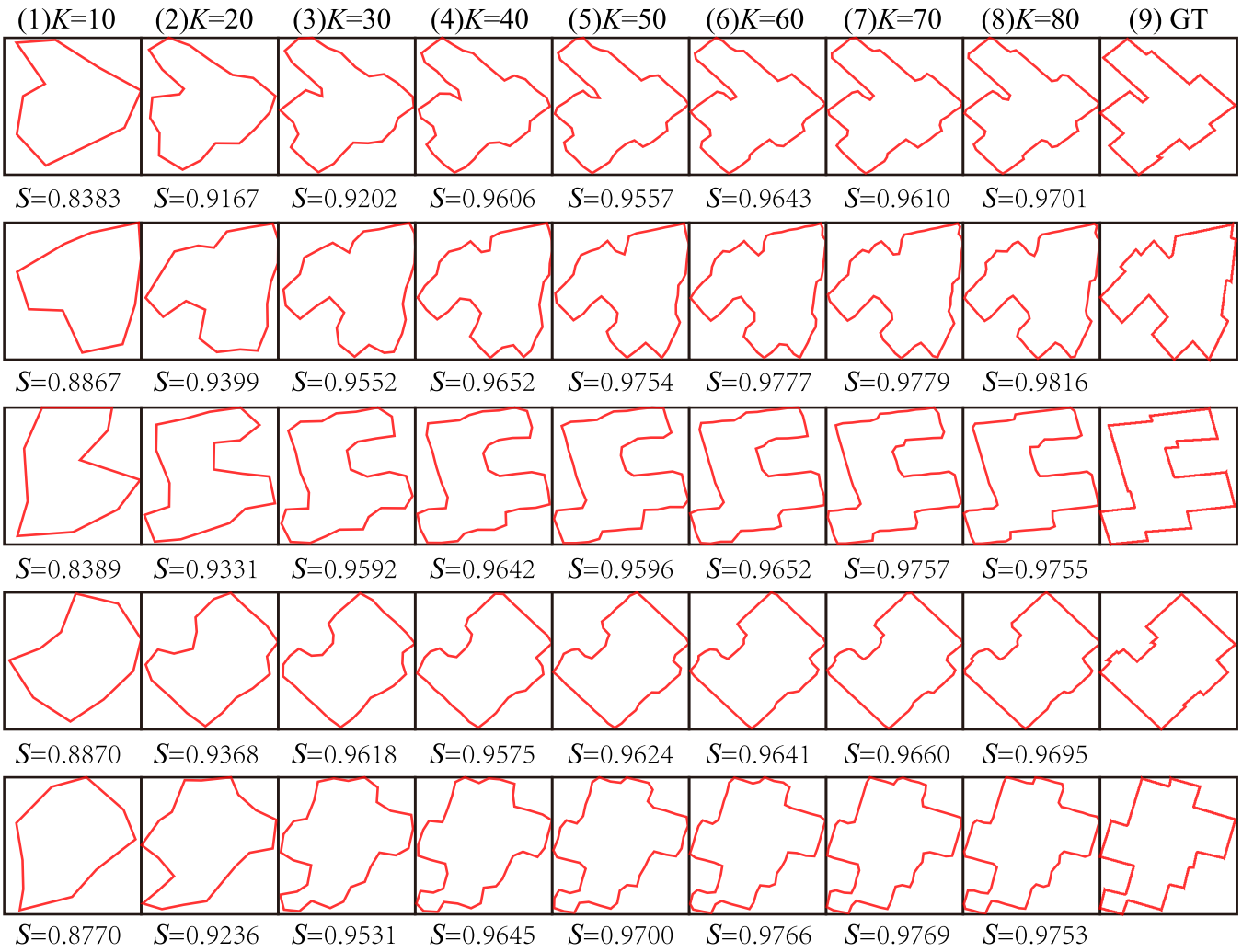


Fig. 15. Relationship between the Fourier shape descriptors' length and approximate accuracy.

number of Fourier descriptors, this study conducted experiments on setting the number of Fourier shape descriptor vectors. This study used the algorithm to verify the ability of the Fourier descriptor length to represent building outlines [45]. This method determines the length of the Fourier shape descriptor vector by using the overlap between the restored building polygon and the original building polygon. The original building polygon is B_o, the fitting region of the Fourier shape descriptor is B_F, the overlapping area of B_o and B_F is $B_{o\cap F}$, and the approximate degree S of the Fourier shape descriptor is shown in the following formula:

$$s = \frac{2 * A(B_{O\cap F})}{A(B_o + A(B_F))} \tag{9}$$

where $A(B_o)$, $A(B_F)$, and $A(B_{o\cap F})$ are the areas of B_o, B_F, and $B_{o\cap F}$, respectively. The larger the S, the stronger the Fourier descriptor's ability to represent building polygons. As the number of vectors in the Fourier descriptor increases, the shape of the polygon fit by the descriptor continues to approach the original polygon. The results of the fit building polygon are shown in Fig. 15.

The Fourier transform-based shape presentation is consistent with human cognition. In Fig. 15, when $K = 80$, the approximation of the Fourier shape descriptor is more significant than 0.97, indicating that the fit polygonal shape has restored the original building shape information. Therefore, this study selected 80 vectors to describe the shape of the building and improved the shape problem in the extracted building outline by fusing the single footprint information extracted by the RNN module with the global Fourier shape descriptor.

VI. Conclusion

Current research on building outline extraction based on remote sensing images has proposed several methods to advance the extraction of vectorized building footprints. However, the extracted vectorized outlines still have numerous shape issues (intersection boundaries, incomplete shapes, and so on). This study introduces the structurally simple PolyMapper as the baseline to mitigate these issues. The proposed transformer block and shape-aware loss function apply shape constraints to the boundary and footprint features of the extracted individual building outlines. The Fourier descriptor provides global constraints for footprint generation in the RNN block by describing global shape features, further solving the problem of shape errors in building outlines. Experimental

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results show that the proposed shape constraint methods solve the shape problem in the current building outline extraction methods. Moreover, this study has more advantages in shape features than state-of-the-art methods. However, this research uses a fixed number of footprint generation methods and, thus, produces footprint redundancy. Model automation will generate an uncertain number of building corners in the future. The dataset used in this study is orthorectified high-resolution remote sensing images, so the shadows and heights between adjacent buildings cannot affect the extraction results. In addition to orthorectified remote sensing images, the extraction of buildings from remote sensing images obtained through oblique photography will involve more influencing factors, such as shadows, occlusion, and different height differences of buildings. The research on refined extraction of buildings in complex scenes will also become a focus of future research.

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